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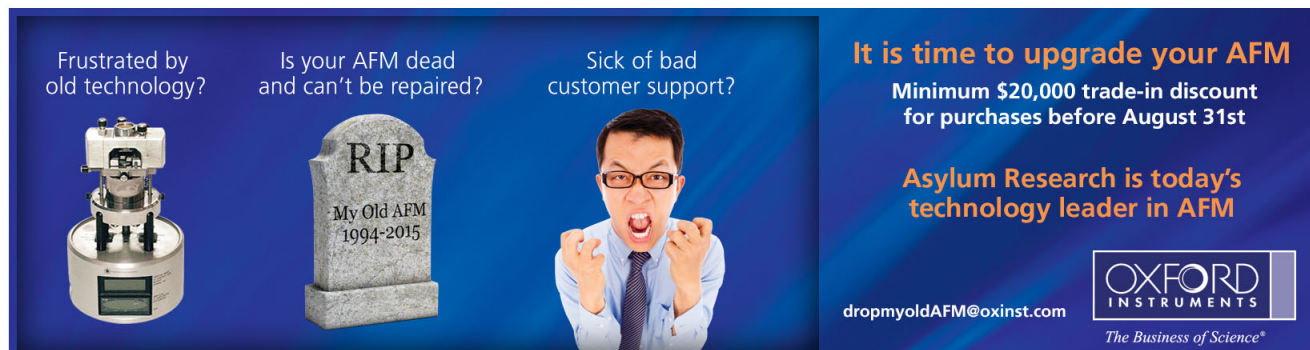
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A comparison of ion transport in different polyethylene oxide–lithium salt composite electrolytes

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Polyethylene oxide (PEO) and lithium salts (LiI, LiClO₄, and LiCF₃SO₃) based composite polymer electrolytes have been prepared for different [EO]/[Li] ratio using solution casting method and characterized by x-ray diffraction (XRD), scanning electron microscope, and differential scanning calorimetry (DSC). The electrical measurements have been carried out at different temperatures. XRD patterns, field emission scanning electron micrographs, and DSC clearly depict that amorphous region in the PEO–LiClO₄ composite electrolyte with [EO]/[Li]=9 is dominant over the other electrolytes. It has been observed that the electrical conductivity of the PEO–LiClO₄ electrolyte is higher than that of the PEO–LiI and PEO–LiCF₃SO₃ composite electrolytes. It has been also observed that the hopping rate of charge carriers for the PEO–LiClO₄ electrolyte is higher than that of the other two electrolytes. The concentration of charge carriers is not thermally activated and the mobility controls the temperature dependence of electrical properties. Scaling of the conductivity spectra has been performed in order to get insight into the relaxation mechanisms.

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I. INTRODUCTION

Over the last two decades polymer electrolyte complexes having considerable ionic conductivity have been studied extensively for application in solid-state lithium ion rechargeable batteries.^{1–5} In particular, PEO–lithium salt complexes^{6–8} are very interesting due to the high energy density, better cycle ability, and safety⁹ when they are used as electrolytes in solid-state battery.

Though PEO has better flexibility, good complexation properties, and good mechanical stability,¹⁰ the conductivity below melting point of PEO is restricted due to high crystalline phase. Some authors^{7,11} have reported that insertion of inorganic salts reduces the crystallinity and, therefore, increases the conductivity of composite electrolytes. It is well established that the nature of polymer–salt complexes, at room temperature, is biphasic consisting of both amorphous region and crystalline phase. Li⁺ ion transport in PEO electrolytes is, generally, governed by the local relaxation and segmental motion of PEO chains. The charge transfer process occurs when PEO is in the amorphous phase.^{11–14} The high dissociation degree of lithium salts is essential for achieving high ionic conductivity in polymer complexes. This can be obtained when lattice energy of the salts is low and the host polymer has high dielectric constant.⁶ Also polymers containing a Lewis base (in case of PEO it is ether oxygen) facilitate the dissolution of the salts due to coordination with the cations. Driven by their potential application in devices, there has been much interest in the mechanism of ion conduction in solid polymer electrolytes.¹⁵

The study of electrical relaxation in polymer–lithium salt complexes provides an important bearing on the viability of

such materials for use as electrolytes in lithium ion batteries, which is still an unresolved problem. In order to gain further insight into the understanding of ion transport processes we have studied the electrical conductivity and relaxation in PEO–LiX (X=I[−], ClO₄[−], CF₃SO₃[−]) composite polymer electrolytes for different [EO]/[Li] ratio.

II. EXPERIMENTAL PROCEDURE

For the preparation of PEO–LiI/LiClO₄/LiCF₃SO₃ composite electrolyte PEO (MW=400 000, Aldrich) and appropriate amount of vacuum dried anhydrous LiI/LiClO₄/LiCF₃SO₃ (Aldrich) were dissolved in acetonitrile under stirring condition. The molar ratio of the ethylene oxide segments to the lithium ions (i.e., [EO]/[Li]) were varied during preparation. After stirring for 8–10 h at room temperature, the solutions became very thick due to solvent evaporation. Finally, they were cast in PTFE containers and dried in vacuum (at a pressure 0.01 mbar) at 60 °C for 36 h to form the homogeneous free standing films. Good free standing films were obtained for [EO]/[Li]=9, 12, 15, and 18.

X-ray diffraction (XRD) patterns of the as-prepared PEO–LiI, PEO–LiClO₄ and PEO–LiCF₃SO₃ samples were recorded in an x-ray diffractometer (Rich-Seifert, model 3000P) using Cu K_α radiation and scan rate of 0.02°/s. The surface morphology of the films was studied using a field emission scanning electron microscope (FE-SEM) (JEOL JSM-6700F). The differential scanning calorimetry (DSC) experiments were performed at a heating scan rate 10 °C/min in Perkin-Elmer DSC-7 in N₂ atmosphere. Electrical measurements such as capacitance and conductance of the samples were measured at different temperatures in an anhydrous environment in the frequency range 10 Hz–2 MHz using a RLC meter (QuadTech, model 7600). A con-

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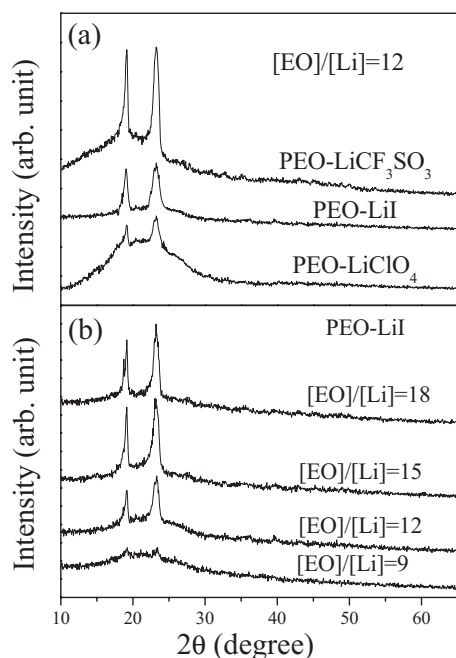


FIG. 1. (a) XRD patterns for PEO–LiClO₄, PEO–LiI, and PEO–LiCF₃SO₃ electrolytes for [EO]/[Li]=12. (b) XRD patterns for PEO–LiI electrolyte for different [EO]/[Li] ratio.

ductivity cell containing two stainless steel blocking electrodes was used for the ac measurements. The dc conductivity at different temperatures was obtained from the complex impedance plots.

III. RESULTS AND DISCUSSIONS

The XRD patterns for the PEO–LiClO₄, PEO–LiI and PEO–LiCF₃SO₃ composite electrolytes are shown in Fig. 1(a) for [EO]/[Li]=12 for each case, while Fig. 1(b) shows the XRD patterns for the PEO–LiI composite electrolyte for various [EO]/[Li] ratio. It is observed that the peak positions occurring due to the crystallinity of the PEO in all the electrolytes remains essentially unaltered but the relative peak intensity of the samples differs. For the PEO–LiClO₄ composite electrolyte [cf. Fig. 1(a)] the peak intensity is the lowest, whereas the peak intensity for the PEO–LiCF₃SO₃ composite electrolyte is the highest. Since the decrease in relative peak intensity corresponds to a decrease in crystallinity, the crystalline phase for the PEO–LiClO₄ composite electrolyte is less than those of the other electrolytes. It is observed in Fig. 1(b) that the peak intensity increases with the increase in [EO]/[Li] ratio for PEO–LiI electrolyte. Other electrolytes show similar behavior for different [EO]/[Li] ratio. Thus the crystalline phase increases with the increase in [EO]/[Li] ratio for all the electrolytes studied here.

Figure 2 shows FE-SEM micrographs for the PEO–LiClO₄, PEO–LiI and PEO–LiCF₃SO₃ composite electrolytes for the same [EO]/[Li] ratio, while Fig. 3 shows FE-SEM micrographs for the PEO–LiClO₄ composite electrolyte for different [EO]/[Li] ratio. The micrographs of all the samples exhibit crystalline structures separated by amorphous region. The small pores, which are present in the micrographs, are caused by the fast evaporation of the acetonitrile solvent during the preparation process. It is further

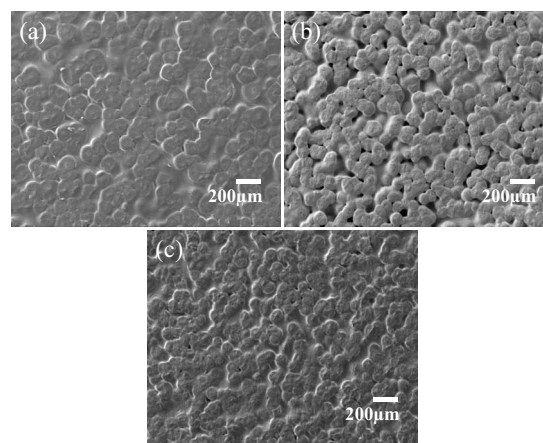


FIG. 2. Field emission scanning electron micrographs for (a) PEO–LiClO₄, (b) PEO–LiI and (c) PEO–LiCF₃SO₃ electrolytes for [EO]/[Li]=12.

observed in these figures that relative smoothness of the surface is better for the PEO–LiClO₄ composite electrolyte than that of other two electrolytes. It is also observed in Fig. 3 that the relative smoothness of the surface decreases and crystalline structures of the PEO becomes prominent when [EO]/[Li] ratio are increased. The smooth surface morphology is closely related to the reduction in the crystallinity of PEO. Thus the crystallinity of the PEO is much less in the PEO–LiClO₄ composite electrolyte for [EO]/[Li]=9, which is also confirmed from XRD studies.

The DSC curves for all the electrolytes are displayed in Fig. 4. The thermodynamic properties such as glass transition temperature (T_g), melting temperature (T_m), and melting enthalpy (ΔH_m) have been obtained from DSC results and are summarized in Table I. The percentage of crystalline PEO (X_c) has been calculated using the equation $X_c = \Delta H_m / \Delta H_{PEO}$, where ΔH_m is the melting enthalpy of the sample and ΔH_{PEO} is the melting enthalpy of completely crystalline PEO, namely 213.7 J/g [Ref. 16]. It is noted in Table I that the percentage of crystalline phase of PEO decreases with the decrease in [EO]/[Li] ratio for each series. It can be observed (Table I) that the effect of the salt concentration ([EO]/[Li] ratio) on T_g for all electrolytes is not sig-

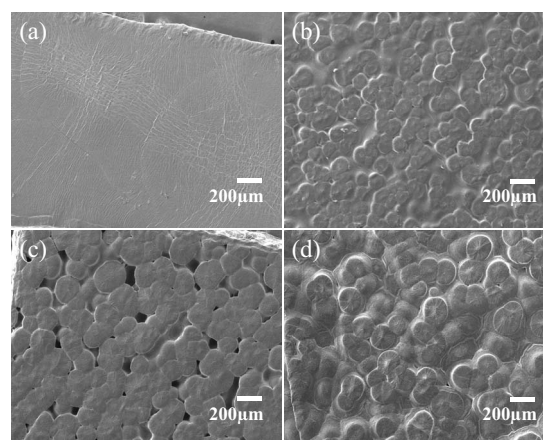


FIG. 3. Field emission scanning electron micrographs for PEO–LiClO₄ electrolytes for (a) [EO]/[Li]=9, (b) [EO]/[Li]=12, (c) [EO]/[Li]=15, and (d) [EO]/[Li]=18.

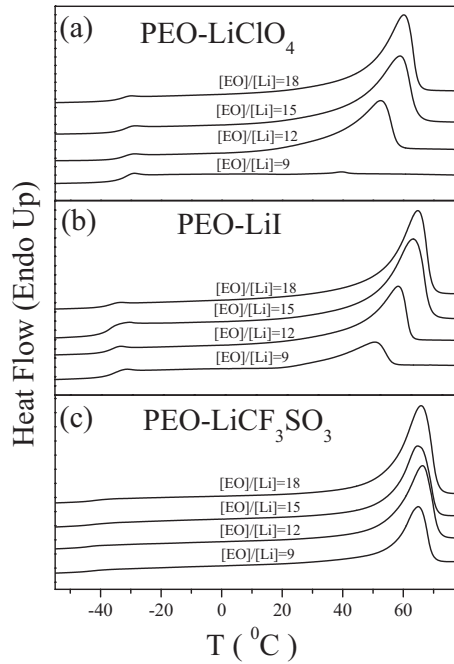


FIG. 4. DSC curves of (a) PEO–LiClO₄, (b) PEO–LiI and (c) PEO–LiCF₃SO₃ electrolytes shown for different [EO]/[Li] ratio.

nificant. One possible reason is that as the salt concentration is increased, formation of ion-pairing occurs and the numbers of free Li ions remain more or less constant.¹⁷ It is also observed that PEO–LiCF₃SO₃ electrolytes possess lowest T_g value. It is worthy to note that LiCF₃SO₃ ions have permanent dipole moments in contrast to LiClO₄ and LiI ions and because of this reason LiCF₃SO₃ ions interact strongly with the polymer chains. This strong interaction together with the complexity of the LiCF₃SO₃ ions decreases T_g for PEO–LiCF₃SO₃ electrolytes.¹⁸ The melting temperature (T_m) shows a gradual increase with the increase in [EO]/[Li] ratio and so also the degree of crystallinity. The amorphous phase present in PEO–LiClO₄ electrolytes is highest as compared with other electrolytes.

The dc electrical conductivity, obtained from the complex impedance plots, is shown in Fig. 5(a) as a function of reciprocal temperature for different electrolytes with the

TABLE I. Comparison of thermal parameters of PEO–LiX (X=I[−], ClO₄[−], CF₃SO₃[−]) electrolytes.

Compositions	[EO]/[Li]	T_g (°C)	T_m (°C)	ΔH (J/g)	%X _c
PEO–LiClO ₄	9	−31.28	39.63	1.986	0.93
	12	−31.56	52.55	48.242	22.57
	15	−31.22	58.96	64.416	30.14
	18	−32.25	60.31	81.810	38.28
PEO–LiI	9	−35.54	48.36	24.664	11.54
	12	−36.35	55.92	51.936	24.30
	15	−35.69	60.64	70.781	33.12
	18	−36.52	62.16	82.108	38.42
PEO–LiCF ₃ SO ₃	9	−42.63	64.82	58.952	27.59
	12	−43.78	66.30	78.168	36.58
	15	−42.57	64.79	88.974	41.64
	18	−41.88	65.96	95.535	44.71

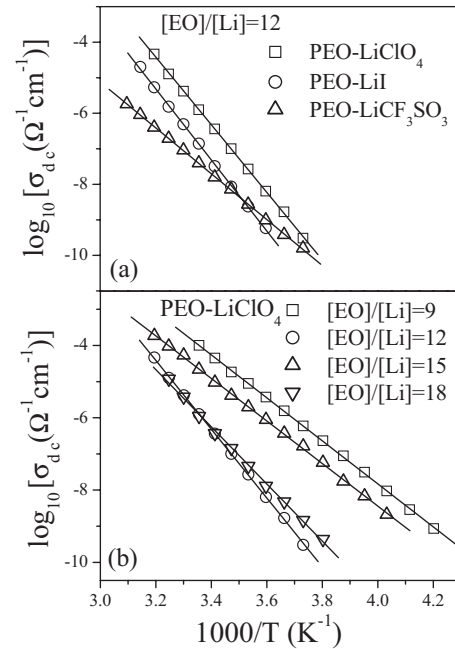


FIG. 5. (a) dc conductivity plotted as a function of inverse temperature for PEO–LiClO₄, PEO–LiI, and PEO–LiCF₃SO₃ for [EO]/[Li]=12. (b) dc conductivity with reciprocal temperature for PEO–LiClO₄ for different values of the [EO]/[Li] ratio. The solid lines are the least square straight-line fits to the experimental data.

same [EO]/[Li] ratio. Figure 5(b) shows the same for the PEO–LiClO₄ composite electrolyte with different [EO]/[Li] ratio. It is noted that the plots in all cases are linear obeying the Arrhenius relation given by

$$\sigma = \sigma_0 \exp(-E_a/k_B T), \quad (1)$$

where, σ_0 is the pre-exponential factor, E_a is the activation energy, k_B is the Boltzmann constant, and T is the absolute temperature. The values of E_a have been obtained from the least square straight-line fits of the data and are shown in Table II. It may be noted that the values of activation energy is very large similar to other polymer electrolytes^{7,19} due to slow hopping processes arising from the strong coupling between mobile cations and the polymer backbone structure.²⁰ It is noted in Fig. 5(a) that at room temperature the conductivity of the PEO–LiClO₄ composite electrolyte is an order of magnitude higher than that for the PEO–LiI composite electrolyte and the conductivity of the PEO–LiCF₃SO₃ composite electrolyte is the lowest. As the temperature is increased the conductivity of the PEO–LiI electrolyte increases much faster than that of the PEO–LiClO₄ electrolyte, whereas for the PEO–LiCF₃SO₃ electrolyte the increase in conductivity with increasing temperature is very slow. Figure 5(b) indicates that the conductivity of the PEO–LiClO₄ electrolyte is the highest for the lowest value of [EO]/[Li] ratio. Similar results have been observed for the PEO–LiI and PEO–LiCF₃SO₃ electrolytes.

The frequency dependence of the ac conductivity at different temperatures is shown in Fig. 6(a) for the PEO–LiClO₄ composite electrolyte for [EO]/[Li]=12, while Fig. 6(b) shows the same at 303 K for the three composites for a fixed [EO]/[Li] ratio. It is observed that at low frequencies the conductivity is independent of frequency and corre-

TABLE II. dc conductivity (σ_{dc}) at 300 K, activation energy (E_a) for the dc conductivity, activation energy (E_c) for the cross over frequency and power law exponent (n) for different compositions of the PEO–LiClO₄, PEO–LiI and PEO–LiCF₃SO₃ composite electrolytes.

Composition	[EO]/[Li]	$\log \sigma_{dc}$ ($\Omega^{-1} \text{ cm}^{-1}$) T=300 K	$E_\sigma (\pm 0.01)$ (eV)	$E_c (\pm 0.02)$ (eV)	n
PEO–LiClO ₄	9	−3.8622	1.18	1.21	0.61
	12	−5.6881	1.88	1.89	0.54
	15	−4.5033	1.20	1.21	0.50
	18	−5.7449	1.57	1.59	0.50
PEO–LiI	9	−4.1468	1.31	1.30	0.62
	12	−6.6305	1.98	2.00	0.57
	15	−5.7789	1.69	1.75	0.50
	18	−5.8269	1.64	1.65	0.52
PEO–LiCF ₃ SO ₃	9	−5.178 41	1.16	1.19	0.58
	12	−7.2962	1.27	1.30	0.54
	15	−6.491 94	1.06	1.05	0.58
	18	−6.225 43	1.18	1.19	0.58

sponds to the dc conductivity for all composites and temperatures. But at higher frequencies there is a cross over from the dc conductivity to the dispersive conductivity, which marks the onset of a relaxation process. The crossover frequency shifts toward higher frequencies as the temperature is increased. It is further observed that the ac conductivity of the PEO–LiClO₄ composite electrolyte is much higher than that of the other electrolytes similar to the dc conductivity.

The ac conductivity of different materials, including polymeric materials, has been studied in the framework of the power law formalisms.^{21–23} In the power law formalism^{21,22} the ac conductivity, $\sigma(\omega)$ is expressed by

$$\sigma(\omega) = \sigma_{dc} [1 + (\omega/\omega_c)^n], \quad (2)$$

where σ_{dc} is the dc conductivity, ω_c is the crossover frequency, and n is the power law exponent, which measures

the interaction between the mobile charge carriers. Such power law conductivity has been observed earlier in glassy and other materials.^{24–26} In general, the value of n is generally less than unity. However, the values of n greater than unity have been observed for some polymeric materials.²⁷ We have applied the above model to study the ac conductivity of the materials under investigation. The conductivity data at different temperatures have been fitted to Eq. (2), using σ_{dc} , ω_c , and n as variable parameters. We have observed that the values of σ_{dc} obtained from the fits match perfectly with those obtained from complex impedance plot (cf. Fig. 5). The variation in the crossover frequency, obtained from the fit, with reciprocal temperature is shown in Fig. 7. It is noted that the crossover frequency is thermally activated, obeying Arrhenius relation

$$\omega_c = \omega_0 \exp(-E_c/K_B T), \quad (3)$$

where E_c is the activation energy for the crossover frequency. The value of E_c has been obtained from the least square straight-line fits of the data and is summarized in Table II. It is observed in Table II that the values of the activation en-

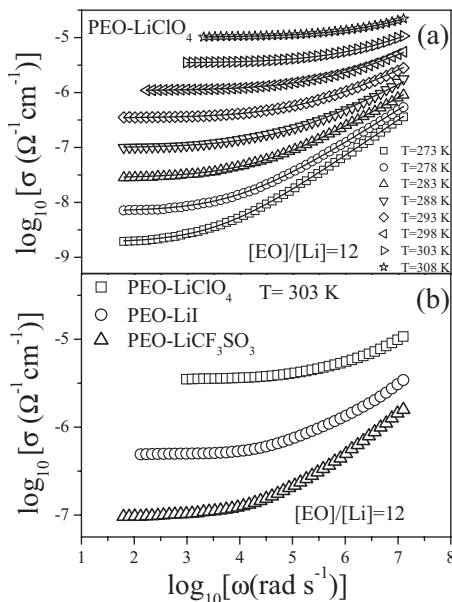


FIG. 6. (a) ac conductivity at different temperatures shown as a function of frequency for PEO–LiClO₄ electrolyte with [EO]/[Li]=12. (b) The ac conductivity at T=303 K shown as a function of frequency for the different electrolytes with same [EO]/[Li]=12.

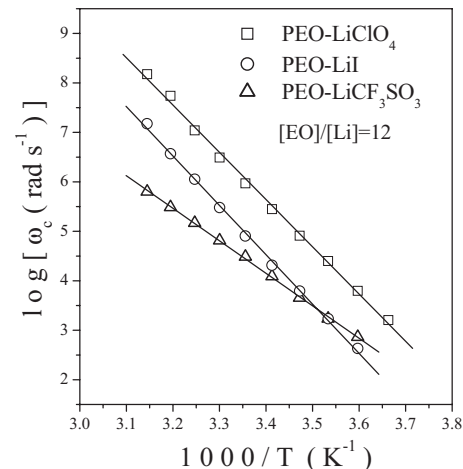


FIG. 7. Dependence of crossover frequency on reciprocal temperature for PEO–LiClO₄, PEO–LiI, and PEO–LiCF₃SO₃ electrolytes having [EO]/[Li]=12. The solid lines are the least square straight-line fits to the experimental data.

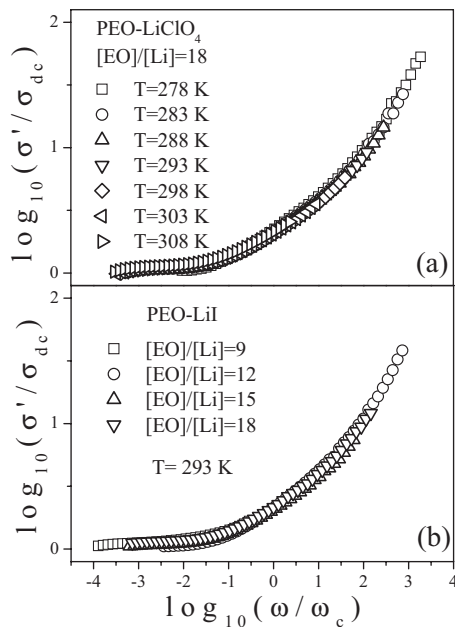


FIG. 8. (a) Scaling of the conductivity spectra with temperature for PEO-LiClO₄ electrolytes with [EO]/[Li]=18 shown in the inset. (b) Scaling of the conductivity spectra at T=293 K for PEO-LiI electrolyte with different [EO]/[Li] ratio shown in the inset.

ergy for the crossover frequency are in good agreement with those for the dc conductivity. The above results suggest that the hopping process with same barrier energy is responsible for the dc conduction and the ac relaxation.

It is noted in Fig. 7 that the value of the crossover frequency is the highest for the PEO-LiClO₄ composite electrolyte. The value of the crossover frequency is also the highest for the lowest value of [EO]/[Li] ratio=9 for the three composite electrolytes. It has been observed that the power law exponent n is almost independent of temperature and is much less than unity similar to glassy and other materials.^{24–26}

The charge carrier concentration n_c is related to the hopping frequency by the following Nernst-Einstein relation:

$$\sigma_{dc} = (q^2 d^2 n_c \omega_h / 12 \pi K_B T), \quad (4)$$

where σ_{dc} is the dc conductivity, q is the charge of the carrier, d is the ion hopping distance, and ω_h is the hopping frequency. It is generally assumed that the crossover frequency is a good approximation for the hopping frequency.²⁶ As the activation energies for σ_{dc} and ω_c are approximately same, Eq. (4) predicts that the charge carrier concentration (n_c) is not thermally activated. Thus the mobility, which is proportional to the hopping frequency, controls the temperature dependence of the conductivity.

The scaling behavior of the conductivity spectra provides an insight into the temperature and composition dependence of the ion dynamics. The scaling of the conductivity spectra has been performed using the procedure outlined elsewhere.²⁸ The results at different temperatures for PEO-LiClO₄ electrolytes are shown in Fig. 8(a). The near-perfect overlap of the spectra at different temperatures indicates that the polymer electrolytes obey the time-temperature superposition principle. The scaling results for different

[EO]/[Li] ratio of PEO-LiI electrolytes at a particular temperature are shown in Fig. 8(b), in which it is noted that all the conductivity spectra of different [EO]/[Li] ratio are nearly scaled to a common master curve. Similar results were also obtained for all other compositions. Thus, the relaxation dynamics of charge carriers in these PEO-Li salt based electrolytes follow a common mechanism throughout the entire temperature and salt concentration ranges.

IV. CONCLUSIONS

The temperature dependence of the electrical conductivity and relaxation behavior for different compositions of the PEO-LiI, PEO-LiClO₄, and PEO-LiCF₃SO₃ composite polymer electrolyte films has been investigated. The XRD, SEM, and DSC studies confirm that the amorphous phase is dominant in the PEO-LiClO₄ composite electrolyte with [EO]/[Li] ratio=9 compared to other electrolytes studied here. The conductivity for this composition is highest of the other two series of electrolytes. The crossover frequency of the PEO-LiClO₄ electrolyte is higher than that of the other two composites for all salt concentrations. The hopping rate of charge carriers together with the percentage of amorphous phase of PEO is very important in controlling the conductivity of the polymer electrolytes. The scaling of the conductivity spectra indicates that relaxation dynamics of charge carriers follow a common mechanism for all temperatures and salt concentrations.

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